CSCI 4810

Final Exam
Programming Part

Keypad Decoding

Given a telephone touch-tone pads *dual tone multiple frequency* (DTMF) sound file, write a Matlab function `dtmf_decoder` to return the key pushed. The function takes a sound file name as its input and returns a number to represent the key pushed.

The DTMF sound files on D2L were recorded with `fs = 8000` Hz. You may use them to verify your codes. Signal sequence is in variable `y`.

Notes: your algorithm MUST use FFT and digital filters to decode DTMF sounds. Command *findpeaks* is NOT allowed.

	1209 Hz	**1336 Hz**	**1477 Hz**
697 Hz	1	2	3
770 Hz	4	5	6
852 Hz	7	8	9
941 Hz		0	

For example, if dial_x.mat contains the DTMF sound generated by pushing key 5, then *dtmf_decoder('dial_x.mat')* should return 5.

Deliverable:

dtfm_cododer.m